

LECTURE 5 • DAY 2, 12:00–12:30

Using graphs and tables to summarise your results

SORT IT Module 3 • Data analysis and interpretation of results • Timor-Leste



**“If your figure needs a paragraph
to explain it, redraw the figure.”**

The only question that matters

- Before you build any table or chart, finish this sentence:
- “This table shows the reader that ...”
- If you cannot finish it in one sentence, you do not know what the table is for — and you are not ready to build it.

Part 1

Tables

Eight things every good table has

The eight parts

#	Part	Why
1	Number and title	What, who, where, when
2	Total N stated	The reader never has to add up your column
3	Rows in a sensible order	By frequency, or natural order — never alphabetical by accident
4	n and % together	“96 (53.3)” — count lets them check, % lets them compare
5	Consistent decimals	One decimal place. “78.75324%” is false precision
6	Headers name the denominator	“n (%) of 160 evaluated”
7	A footnote per exclusion	This is where honesty lives
8	No vertical lines	Three horizontal rules. Journal standard

Before — what most first drafts look like

Outcome	Number	Percent
Cured	96	53.33333
Treatment completed	30	16.66667
Died	14	7.77778
Lost to follow up	16	8.88889
Failed	4	2.22222
Not evaluated	12	6.66667
	180	100

After — publishable

Treatment outcome	n (%)
Cured	96 (53.3)
Treatment completed	30 (16.7)
Treatment success	126 (70.0)
Died	14 (7.8)
Lost to follow-up	16 (8.9)
Treatment failed	4 (2.2)
Not evaluated (transferred out)	12 (6.7)
Still on treatment	8 (4.4)
Total	180 (100.0)

Part 2

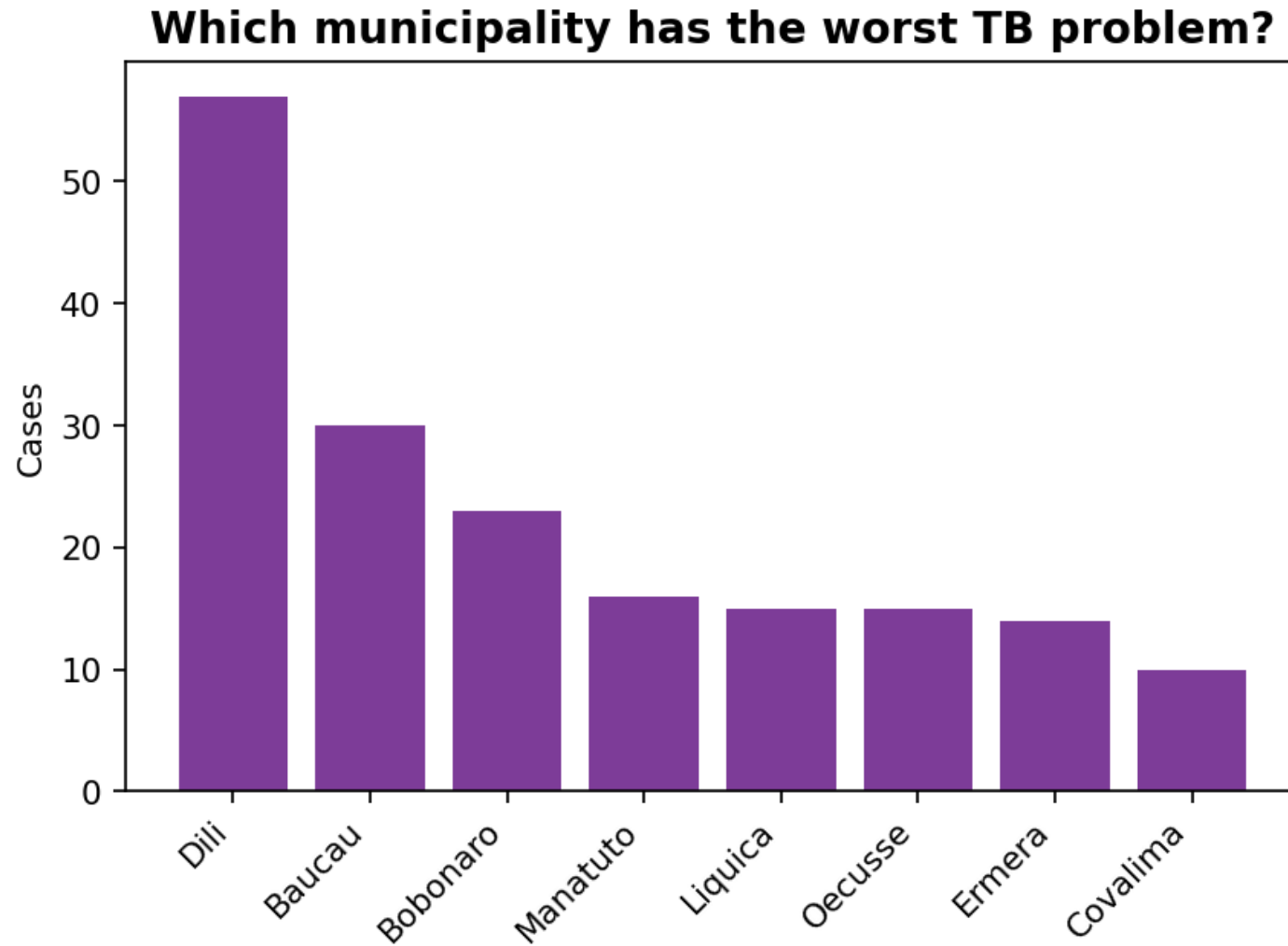
Graphs

Pick the chart that answers your question

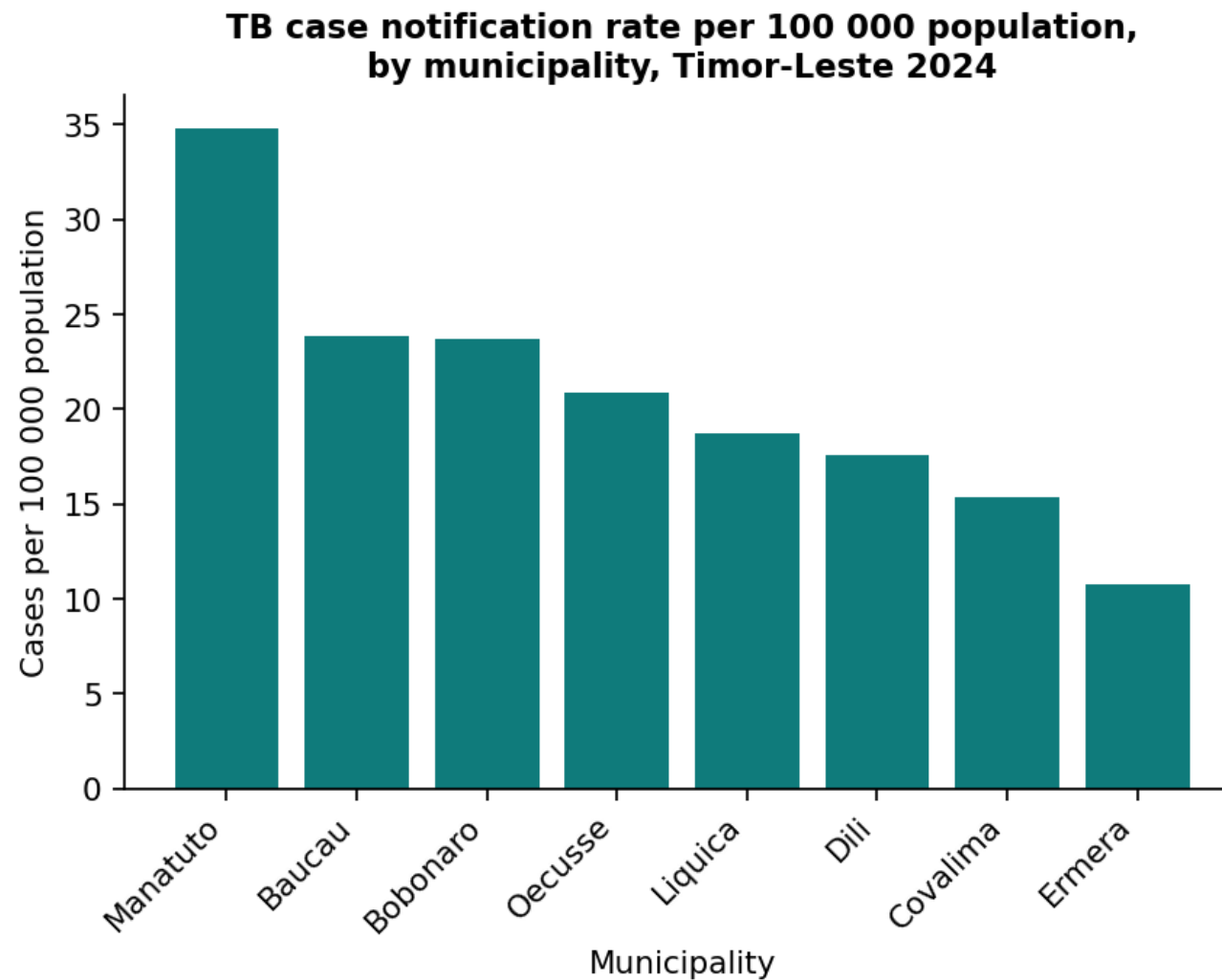
Which chart? Read across.

My question is...	Use	Not
How many in each category?	Bar chart	Pie chart
How is one continuous variable spread?	Histogram	Bar chart
How does it change over time?	Line chart	Bar chart
How do groups compare on a distribution?	Box plot	Bar of means
Is there a relationship between two continuous variables?	Scatter plot	Line chart
What proportion of a whole, across groups?	Stacked bar	Several pies
Where does my cohort lose people?	Flow diagram	Table

Which municipality has the worst TB problem?



Now per 100 000 population.



The four ways a graph lies

Common crimes

- Truncated y-axis — exaggerates small differences
- Counts where you need rates
- Pie chart with nine slices
- Bar of means hiding the spread

The five checks

- Both axes labelled, with units
- y-axis starts at zero
- n is stated on the chart
- Survives black and white printing
- All decoration deleted

AFTER LUNCH • 35 minutes

Chart Doctor

1. Six sick charts are taped around the room
2. Your team diagnoses each one: what is wrong?
3. Then prescribe: what would you do instead?
4. One sticky note per chart. Prize for the best diagnosis.

A table gives exact values.
A graph shows shape and comparison.

Choose one. Never show the same numbers as both.